Tracking the 2025 NBA Finals with YOLOv8

An exploratory computer-vision pipeline on Game 1 highlights: what person detection alone can and cannot recover from broadcast video, and what that teaches about turning pixels into data.

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ABSTRACT A five-stage pipeline ran an off-the-shelf YOLOv8 detector over about twelve minutes of 2025 NBA Finals Game 1 highlights, logging 236,658 person boxes. It then built possessions, shot locations, an expected-field-goal (xFG) model and movement zones from those boxes alone. Re-running each stage on the saved outputs shows where it holds and where it breaks. Detection is plentiful but low-confidence: 30% of boxes clear the 0.6 threshold. The possession heuristic yields 3,352 segments, 1,307 of them zero frames long. The xFG model scores 0.997 AUC only because it is fitted to labels generated by a rule from its own two inputs. The pipeline's value is as a map of what video-to-data work needs: calibration, identities, ground truth and out-of-sample tests.

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Summary

This is a case study of a computer-vision prototype, not a basketball finding and not an investment result. The question it answers is practical: given only a general-purpose person detector and broadcast highlights, how much structured data can be recovered, and how would one know whether it is right?

236,658	30.0%	1,307	0.997
Person boxes logged across 21,933 frames (≈ 12.2 minutes)	Share of boxes above the 0.6 confidence threshold used downstream	Of 3,352 "possessions" that are zero frames long	In-sample AUC of the xFG model on labels made by a rule

KEY FINDINGS

1. **Detection is the strong stage.** YOLOv8n found a median of 11 people per frame. Its confidence is low, though: a median of 0.46, and only 70,995 boxes above 0.6. Frames with any confident box average 3.4 of them.
2. **The possession heuristic over-segments.** 98.3% of segment breaks come from frame-to-frame jumps in the spread of detected people, not from gaps in time. Only 85 of 3,352 segments last a second or more.
3. **The shot model learns its own label.** "Makes" are assigned by a rule, *distance < 20 and defender > 3*, applied to the same two features the model is then fitted on. Its near-perfect separation measures the rule, not shooting.
4. **Pixels are not court feet.** Without camera calibration, "zones" and "distances" describe positions on the screen. The movement zones partition the frame, not the floor.

Each finding below comes from re-running the repository's own rules on its saved outputs, with the reproduced results checked against the stored files. No model was re-run and no video was used for this report. The video is not in the repository, and no broadcast frames are reproduced here.

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Material and scope

The repository holds four pieces of work at different stages. This report concerns the first; the others are described for completeness.

Table 1. Contents of the repository.

Folder	What it is	State
nba_cv_2025_finals	Detection → possessions → shot chart → xFG model → movement zones, on Game 1 highlights	Complete exploratory run; all five stages produced saved outputs
game7	A second, rougher pipeline on a separate game video, sampled every 6th frame at 1280×720	Work in progress. Player identities are placeholders (box index mod 10), and segmentation yields only 7 possessions
Sloane	Scaffold for a detection-to-features pipeline (centroid, speed, acceleration)	One script, never run; other folders empty
2periodconsumption.html	An interactive two-period consumption–saving model	Unrelated to computer vision; out of scope

The clip

Detection starts 88 seconds into a highlights video of the 2025 Finals Game 1. The first logged frame, 2,637, matches a frame rate of 29.97 per second. Boxes run to frame 24,577, a span of 21,941 frames or about 12.2 minutes. Box coordinates reach 1,920 by 1,080 pixels, so the source is 1080p. Because it is a highlights reel, the footage is a sequence of edited plays with camera cuts, replays and crowd shots, not continuous game time. The repository records no ground truth of any kind: no tracking data, play-by-play, shot outcomes or hand counts.

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The pipeline

Every stage after detection works only from the logged boxes. There is no ball detection, no team or role classification, and no identity tracking across frames.

Table 2. The five stages and the rule each applies.

Stage	Script	Rule
Detection	detect_and_log.py	Ultralytics YOLOv8n (pinned to 8.1.14), default thresholds, every frame. Keeps class “person” only, which includes players, referees, coaches and spectators.
Possessions	possession_tracker.py	Uses boxes with confidence > 0.6 and frames with ≥ 2 people. Spread is the mean distance of box centres from their mean. A new segment starts on a gap of more than 15 frames or a spread change of more than 100 px.
Shot chart	shot_chart.py	In every frame, the “shooter” is the confident box farthest from an assumed hoop pixel (960, 1050), mapped linearly to a 50 × 47 grid.
xFG model	shot_difficulty_model.py	Every 20th confident frame is a candidate shot. Features: shooter-to-hoop distance and nearest-other-person distance, in pixels × 50/1,920. The label is simulated by rule. Logistic regression, fitted and scored on the same rows.
Movement zones	heatmap_generator.py	Confident box centres rescaled to 50 × 47. Gaussian KDE heatmap and k-means with k = 6 (unseeded).

The two rules that matter most for interpretation are the segment break and the shot label:

$$new\ segment_t \Leftrightarrow (frame_t - frame_{t-1} > 15) \vee (|spread_t - spread_{t-1}| > 100\ px) \tag{1}$$

$$make = 1 \Leftrightarrow (shot\ distance < 20) \wedge (defender\ distance > 3) \tag{2}$$

A code comment marks rule (2) as a stand-in until real shot results are available.

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Results

Detection

The detector logged 236,658 person boxes in 21,933 of the 21,941 frames, a mean of 10.8 and a median of 11 per frame, peaking at 30. Most are low-confidence: the mean score is 0.496 and the median 0.462. The downstream stages keep only the 70,995 boxes (30.0%) scoring above 0.6. A confident box appears in 20,649 frames, and 17,360 frames have two or more. Where a frame has any confident box, it has 3.4 on average.

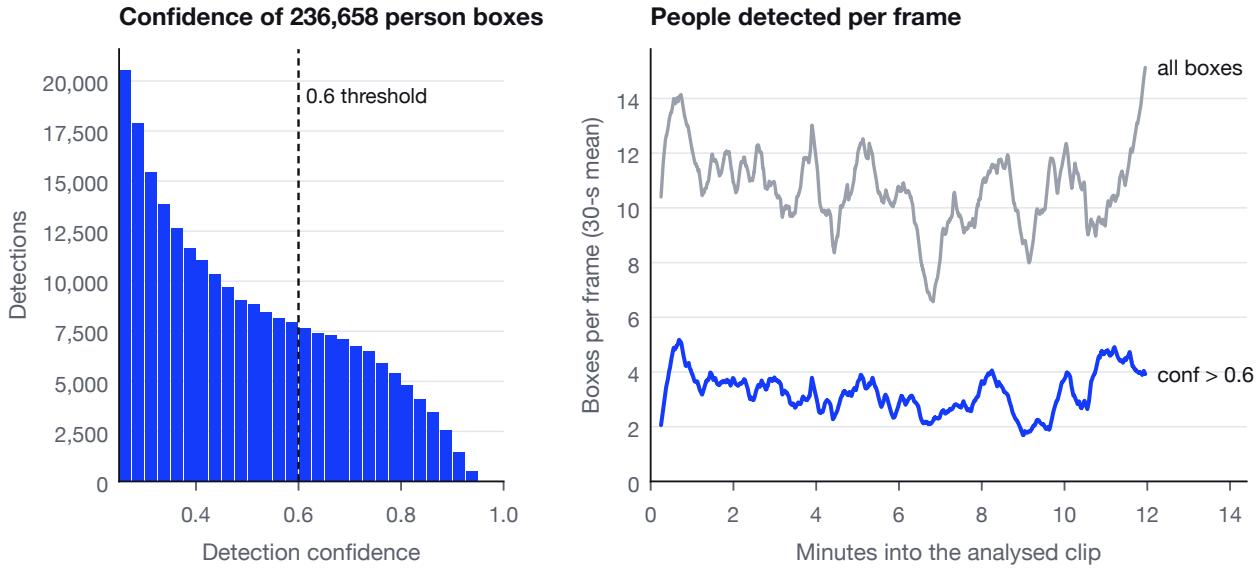


Figure 1. Left: confidence of all person boxes; the downstream threshold is dashed. Right: boxes per frame through the clip, as a 900-frame (≈ 30 s) moving average, for all boxes and for those above 0.6.

Source: computervision/nba_cv_2025_finals/game1_detections.csv; derived for this report.

The gap between the two lines in Figure 1 is the pipeline's first trade-off. The 0.6 cut keeps boxes the model is sure of, but it discards most of the people in each frame, so every later stage reasons about a small, shifting subset of the scene.

Where people appear

The movement stage rescales box centres to a 50 × 47 grid labelled as half-court feet. With no camera calibration, that grid is the screen. Figure 2 shows the density of confident detections in those coordinates, with the six k-means centres the pipeline saved.

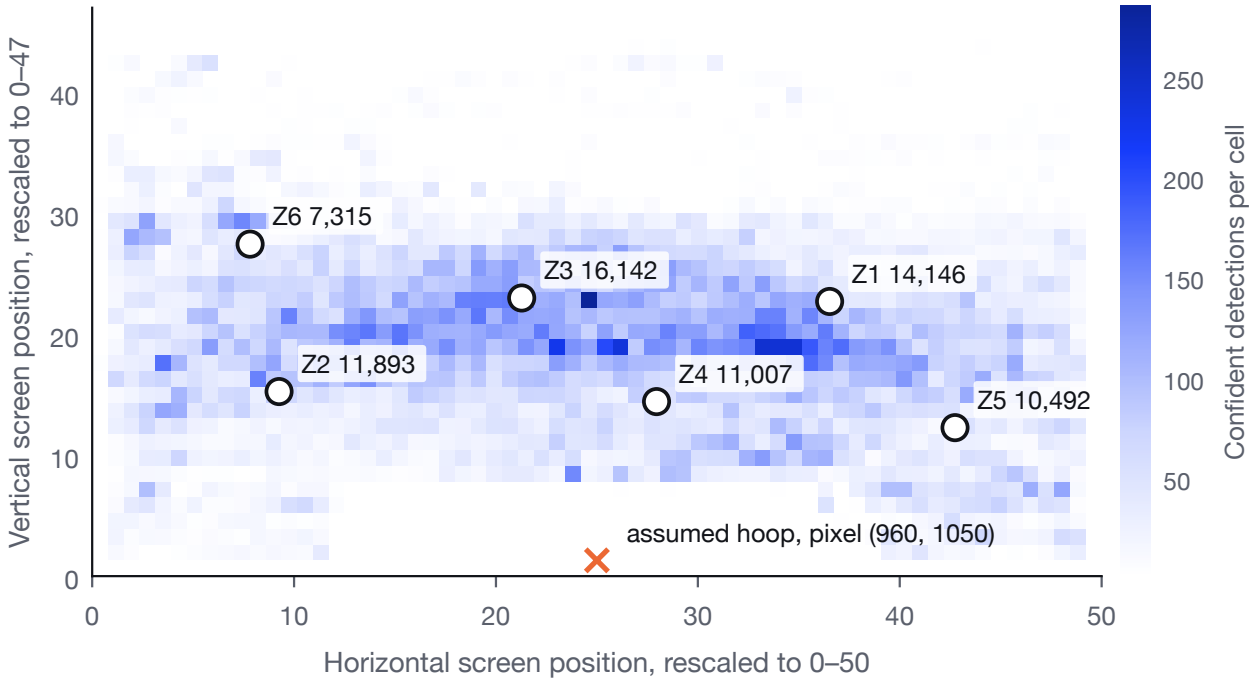


Figure 2. Density of the 70,995 confident box centres in rescaled screen coordinates, with the six saved k-means centres (Z1–Z6, labelled with point counts) and the assumed hoop position. Zone numbering is arbitrary because the clustering was unseeded.
Source: game1_detections.csv; movement_zone_summary.csv.

The zones carve the cloud into six contiguous regions holding between 7,315 (10.3%) and 16,142 (22.7%) of the points. Each has a standard deviation of 3.7–4.9 units on each axis. That is what k-means does to any dense cloud. Because the camera pans and zooms, the same screen position maps to different places on the floor from shot to shot, so the zones say little about where players were. The assumed hoop pixel sits near the bottom edge of the frame. Whether a hoop is ever there depends on the camera angle, which changes with every cut.

Possessions

Re-running rule (1) on the confident boxes reproduces the saved table exactly: 3,352 segments with identical start frames and lengths. Their lengths are short. The median is one frame, 1,307 segments (39.0%) have zero length, and only 85 last a second or more. Six last three seconds or more, the longest 258 frames (8.6 s).

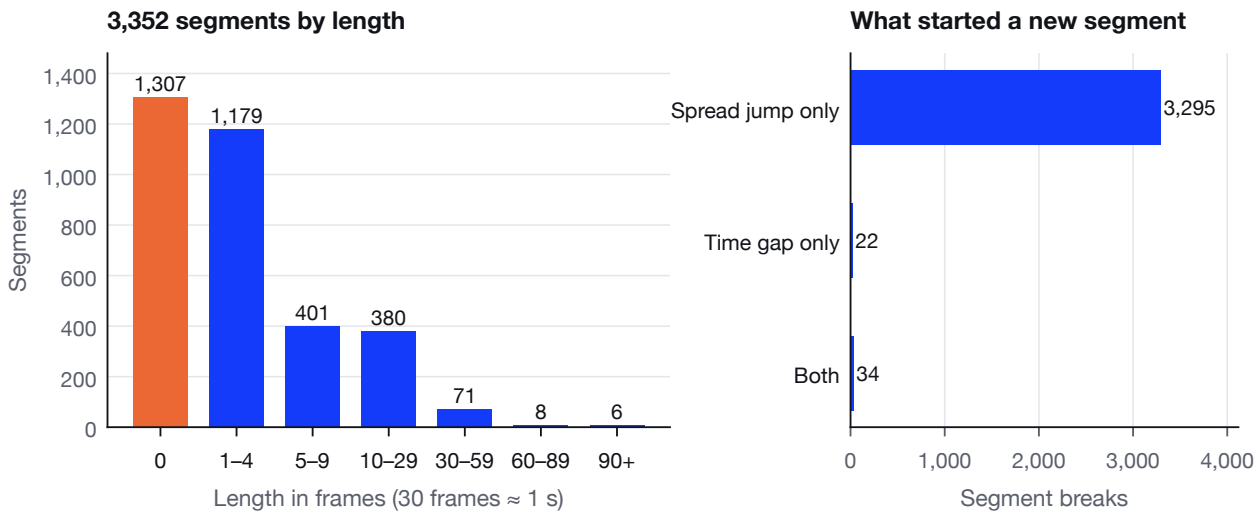


Figure 3. Left: the 3,352 segments by length in frames; zero-length segments are highlighted. Right: which part of rule (1) triggered each of the 3,351 breaks.

Source: possessions_summary.csv; rule re-run on game1_detections.csv and checked against it.

The right panel explains the left. Of 3,351 breaks, 3,295 (98.3%) are triggered by the spread condition alone, against 22 by time gaps alone and 34 by both. The spread of the confident subset is noisy, with a median frame-to-frame change of 25.4 px and a mean of 56.1 px. It jumps whenever a box crosses the 0.6 threshold or a new face enters the frame, so a 100 px threshold fires constantly. A plausible reading, not tested here, is that most breaks reflect detection flicker and camera cuts, not changes of possession.

Shots and expected field goals

Taking every 20th confident frame gives 1,033 candidate “shots”. Of these, 881 have at least two people and are kept. The shooter proxy is the person farthest from the assumed hoop pixel, which caps the distance proxy at 37.0 units; observed values run from 9.0 to 34.4, with a mean of 21.7. Rule (2) labels 284 of the 881 (32.2%) as makes, and the saved labels match the rule on every row.

A logistic regression on those two features separates the labels almost perfectly (Figure 4): in-sample AUC 0.997 and 98.4% accuracy at a 0.5 cut-off. Mean xFG is 0.90 for labelled makes and 0.05 for misses. Its mean across all rows, 0.322, simply reproduces the label rate. None of this says anything about shooting, because the label is a deterministic function of the inputs and the model is scored on the rows it was fitted to. The defender condition barely binds: only 12 candidates (1.4%) have another person within 3 units.

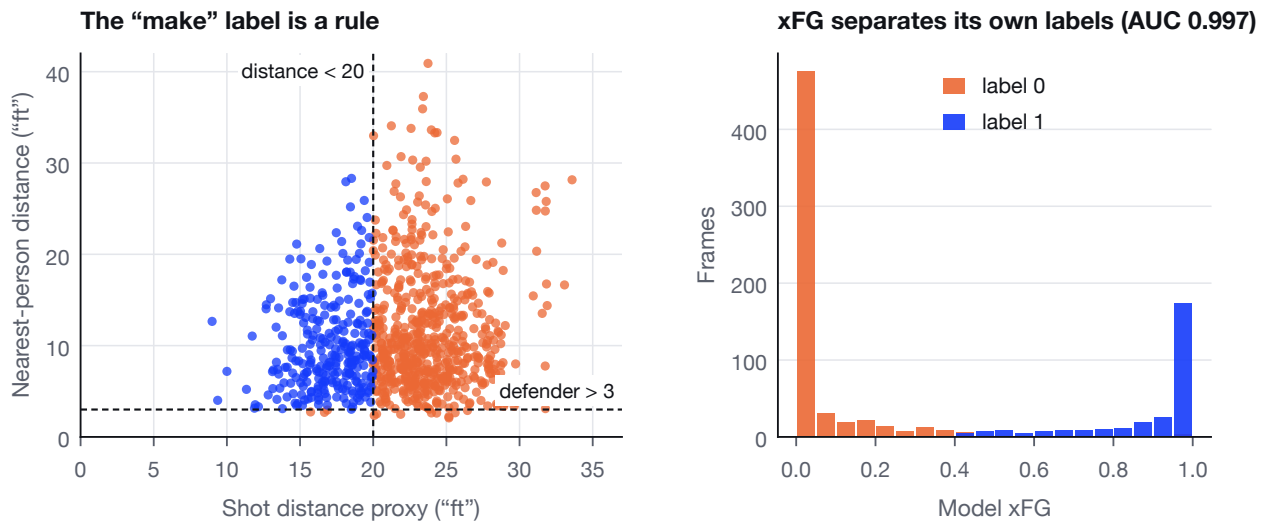


Figure 4. Left: the 881 candidate shots in the model’s two features, coloured by the simulated label (blue 1, orange 0, as in the right-hand legend); dashed lines are the thresholds of rule (2). Right: the fitted xFG for each label.

Source: shot_difficulty_output.csv; AUC and accuracy derived for this report.

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What the numbers cannot tell you

- **No ground truth.** There is nothing to score detections, possessions, shooters or makes against, so every rate in this report describes the pipeline, not the game.
- **No identities or roles.** “Person” covers players, referees, coaches and crowd. Without tracking IDs, the “shooter” and “defender” are just whichever boxes satisfy a distance rule in a single frame.
- **No calibration.** Pixel positions are rescaled linearly and labelled as feet. The scripts even disagree on frame height: the shot chart assumes 720 pixels, the other stages 1,080. Distances and zones are screen measurements.
- **Edited footage.** Highlights splice plays, replays and cutaways. A heuristic that reads continuity from frame to frame will treat each cut as an event.
- **Simulated, in-sample labels.** The xFG model is fitted and evaluated on labels generated from its own features, with no held-out data.
- **One clip, one run.** The k-means step is unseeded and the video is not in the repository, so the detections cannot be regenerated from source.

The game7 pipeline shares these limits and adds one: its speeds come from placeholder identities. 93.8% of its 28,859 feature rows are classed as “fast”, an artefact of measuring speed between unrelated boxes.

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From the court to alternative data

The same chain of steps, detecting objects frame by frame, aggregating them into counts and positions, segmenting time, then modelling, is the general shape of turning imagery into a dataset, whatever the subject. This project does not test any such use. What carries over is a checklist, and each item is a gap this pipeline exposes:

1. **Fix or calibrate the camera before measuring.** A count or position means something only if the field of view is stable, or mapped to real coordinates with a homography. Moving broadcast cameras break both.
2. **Measure the detector before trusting the count.** Hand-label a sample of frames and report recall and precision at the chosen threshold. Here, 70% of boxes fall below the cut-off and their accuracy is unknown.
3. **Filter to the objects that matter.** A class such as “person” mixes the subjects of interest with bystanders. Role or region filters belong before aggregation, not after.
4. **Track identities when the question is about behaviour.** Speeds, dwell times and paths require the same object across frames. Per-frame boxes cannot provide them.
5. **Use labels that are independent of the features.** A target must come from outside the inputs, or the model can only rediscover the rule that made it.
6. **Validate out of sample and over time.** Fit on one period or location and test on another before reading anything into accuracy.

As built, the pipeline connects detection, aggregation and modelling end to end. Calibration, identity tracking and ground truth, the work that would make its outputs usable as data, remain to be done.

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Data and reproducibility

REPOSITORY	ajaiupadhyaya/computervision, commit 740e6c4, 14 Aug 2026. The pipeline and its outputs were committed in July 2025.
DETECTIONS	nba_cv_2025_finals/game1_detections.csv: 236,658 rows (frame, label, conf, x1, y1, x2, y2), frames 2,637–24,577. The source video and model weights are not in the repository.
STAGE OUTPUTS	possessions_summary.csv (3,352 rows), shot_difficulty_output.csv (881 rows), movement_zone_summary.csv (6 zones, 70,995 points).
THIS REPORT	All four figures were drawn from the saved CSVs. The possession rule was re-run on the detections and asserted equal to the saved table, and the make labels were asserted equal to rule (2) on every row. Break triggers, length bins, confidence and per-frame statistics, AUC, accuracy and zone shares are derived here from the same files. Detection used ultralytics 8.1.14 (requirements.txt).

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